Vineet Vora

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Summary

MS Artificial Intelligence student building open-source AI systems, model-serving workflows and distributed data pipelines. Hands-on experience with vLLM inference on A100 GPUs, Hadoop on AWS and Linux HPC cluster setup. Owns projects from scoping and implementation through evaluation and delivery.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript (ES6+), SQL
AI / ML: PyTorch, HuggingFace Transformers, vLLM, LoRA / PEFT, TensorFlow, Scikit-learn, OpenCV
Data & Infrastructure: Hadoop, MapReduce, HDFS, YARN, AWS EC2, SLURM, Docker, CUDA, PostgreSQL, MongoDB
Web & Delivery: React.js, Node.js, Cloudflare Pages, Git, CI/CD, scoping, Design Thinking (IBM)

Experience

New Jersey Institute of Technology — Student Intern, HPC *Sep 2026 – Present*
- Build and test a Raspberry Pi mini HPC cluster; install Red Hat, Slurm and supporting software.
- Troubleshoot compute-node connectivity and remote access using SSH, network diagnostics and head/compute-node checks.

Selected Work

TRIDENT SQL — Natural Language → SQL over Enterprise Databases (MS Thesis) *Jan 2026 – Present*
github.com/VineetV2/Trident-SQL | Python, vLLM, PyTorch, gpt-oss-120b, SLURM, SQLite
- Built a **four-phase, training-free text-to-SQL pipeline** and designed its **Plan-and-Fragment** SQL writer, which builds each query from exact SQL fragments, plus **Cross-Writer Agreement** voting with an LLM judge for close calls.
- Designed a **Query Tree** check: it turns the question and the generated SQL into trees, compares them branch by branch, and repairs only the flagged branches.
- Achieved **70% execution accuracy on BIRD dev** (1,534 questions, 11 real-world databases) using **only open-weight models and no proprietary API** — so it runs fully inside a customer's own environment with no data leaving it.
- Served gpt-oss-120b with vLLM on NJIT's Wulver SLURM GPU cluster, with checkpoint/resume so an 8-hour run never restarts from zero; compared four open backbones, then cut five selection variants to one.

Geomagnetic Storm Forecasting from NASA Solar Imagery *Dec 2025 – Present*
github.com/VineetV2/imf-bz-forecasting-surya | Python, PyTorch, LoRA, CUDA, Weights & Biases
- Adapted Surya, a 366M-parameter Vision Transformer pretrained on NASA SDO imagery, to forecast interplanetary magnetic field Bz at +24h/48h/72h — what utilities watch before a storm.
- Compared LoRA against a frozen encoder on 13-channel AIA EUV and HMI inputs (512×512); the frozen encoder reached **3.39 nT RMSE** on **0.34% of parameters**, **3.5**× more efficient than LoRA.
- Validated by Leave-One-Flare-Out cross-validation across 57 M5.0+ and X-class flares; 96.5% of folds succeeded.

Distributed Flight Analytics — 118.9M Records on Hadoop *Nov – Dec 2025*
github.com/VineetV2/hadoop-flight-data-analysis | Java, Hadoop MapReduce, AWS EC2
- Stood up a 6-node Hadoop cluster on AWS EC2 and processed 22 years of US flight data — **118.9M records, 11.66 GB** — reaching **100% data-local map execution** and **2.73× speedup** under horizontal scaling.

Peon — Local-First Memory System for AI Coding Agents *2026*
github.com/VineetV2/peon-mem | TypeScript, MCP, SQLite
- Shipped an MCP server giving AI coding agents persistent project memory: automatic capture, LLM consolidation, hybrid retrieval, daily self-audit. Drops into any MCP client.

Leadership

VITOPIA, VIT-AP University Flagship Festival — Head of Sponsorship & Core Team *Jan – Feb 2024*
- Led a **20-person team** through outreach to **450+ companies**, closing **$180,000** (INR 1.5 Cr) in corporate sponsorship and exceeding every revenue target.
- Owned the commercial relationship end to end — pitch, negotiation and partner management with non-technical stakeholders — alongside vendor, venue and scheduling logistics.

Beat The Heat, VIT-AP — Club Administrator & Head of Finance *Aug 2022 – Jan 2025*
- Ran the club's annual budget and supplier negotiations for two and a half years, setting the financial strategy the committee planned expansion against.

Publication

Preventing Wildfires in Energy Transmission by Automatic Power Line Defects Detection Using Machine Learning and AI
2025 New Jersey Future Energy Transmission Conference (NJFET), IEEE — Dec 2025 | presented | DOI 10.1109/NJFET67489.2025.11380637
- Computer vision framework moving utilities from reactive repair to preventive maintenance on power line defects and vegetation encroachment in UAV imagery; built the falling-wire proof of concept with Canny edge detection and a Probabilistic Hough Transform.

Education

New Jersey Institute of Technology — MS, Artificial Intelligence | GPA 3.83 | Newark, NJ *Expected May 2027*
Advanced NLP, Computer Vision, Information Retrieval, ML for Data Science, Big Data
Vellore Institute of Technology – AP — BTech, Computer Science | India *Jun 2025*